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Enhancing knowledge base question answering with AMR-driven subgraph retrieval

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ABSTRACT

Knowledge Base Question Answering (KBQA) aims to answer natural language questions using factual triples from a knowledge graph (KG). Mainstream Graph Neural Network (GNN)-based methods rely on subgraph retrieval to reduce reasoning complexity. However, existing retrieval techniques, which predominantly depend on shallow similarity matching between question text and KG relations, often fail to bridge the semantic gap between linguistic diversity and structured knowledge. This leads to inconsistent subgraph quality and poor interpretability, thereby hindering QA performance. To address this fundamental challenge, we propose a novel framework driven by Abstract Meaning Representation (AMR). Our method parses questions into AMR graphs that preserve core semantics, from which we extract canonical predicate-argument structures to serve as high-precision filters for pruning and refining an initially retrieved noisy subgraph. Furthermore, we leverage the AMR structure to inform the GNN's adjacency matrix and to generate highly interpretable reasoning chains. Experiments on the WebQSP and CWQ benchmarks demonstrate that our approach not only significantly improves subgraph quality and retrieval performance but also achieves competitive KBQA results.

1. Introduction

Knowledge graphs (KGs) represent factual knowledge in a structured format using triples (*head*, *relation*, *tail*), capturing entities and their interrelations (Paulheim, 2016). Knowledge Base Question Answering (KBQA) aims to identify answer entities within a KG based on a natural language question. This task can be effectively framed as a node classification problem, where entities in the KG are classified as either answers or non-answers for a given query (Mavromatis & Karypis, 2025).

While large language models (LLMs) excel in many natural language processing (NLP) tasks, they encounter significant challenges with complex reasoning and specialized domains, often producing hallucinated or inaccurate outputs (Zhang et al., 2025). In contrast, graph neural networks (GNNs) have gained prominence in KBQA systems due to their inherent capability to model and reason over intricate graph structures (Schlichtkrull et al., 2018). However, directly applying GNNs to reason over an entire large-scale KG, which can often contain billions of triples, is computationally infeasible due to prohibitive memory and processing

requirements. Consequently, subgraph retrieval becomes an essential preprocessing step to isolate a relevant portion of the KG for subsequent reasoning.

Existing subgraph retrieval methods predominantly rely on measuring the similarity between the question and KG relations using text embeddings. Entities and paths are then selected based on these similarity scores. For example, PullNet (Sun et al., 2019) employs an LSTM-based retriever to iteratively select relations relevant to the question, followed by a GNN-based reasoner to identify the answer tail entities. Similarly, SR (Zhang et al., 2022) decouples the retriever and reasoner, constructing subgraphs from the top-k retrieved paths. However, a key limitation of these embedding-based approaches is their frequent inability to adequately capture the nuanced semantic structure of the question. This often results in retrieved subgraphs containing noisy nodes and irrelevant relations, which can impair downstream reasoning performance (Jain et al., 2021).

A natural language question often consists of multiple relations that can serve as explicit guidance for subgraph retrieval. Abstract Meaning

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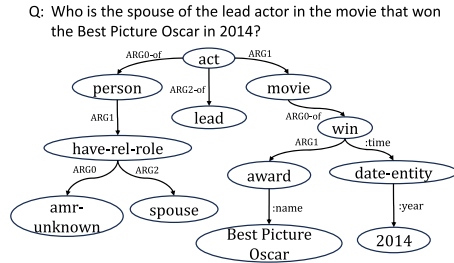


Fig. 1. An example of utilizing AMR to extract semantic information from a complex question.

Representation (AMR) provides a structured representation of a question’s semantic information, stripping away syntactic variations while preserving the core relations (Banarescu et al., 2013). As illustrated in Fig. 1, to solve this problem, the first step is to identify the film that won the Best Picture Oscar in 2014. The next step is to determine the lead actors of the film, followed by identifying the name of the spouse as the final answer. This process corresponds to the predicates in the AMR graph, such as *win*, *act*, and *spouse*. This clear graph structure enhances the quality of subgraph retrieval and provides interpretability for the reasoning process.

Compared to raw text input, an AMR graph captures more fine-grained semantic information and presents it in a graph structure analogous to a knowledge graph. Motivated by these advantages, we propose an AMR-based subgraph retrieval method for KBQA with the following objectives: (a) Transform the question into an AMR graph to capture richer semantic information. (b) Extract fine-grained relations based on nodes and edges in the AMR graph. (c) Leverage the extracted relations to guide subgraph retrieval, to ensure compact and precise subgraphs. (d) Integrate relations extracted from AMR into the adjacency matrix for improving GNN-based reasoner. Subsequently, a subgraph-oriented reasoning model is employed to identify answer entities while ensuring an interpretable reasoning process.

Our key contributions can be summarized as follows:

- **A Novel AMR-driven Subgraph Refinement Framework.** We propose a novel framework that, unlike prior works using AMR to generate full logical forms, uniquely employs AMR as a semantic bridge to refine and prune noisy subgraphs generated by general-purpose retrievers. This directly targets the semantic gap problem in existing KBQA pipelines.
- **Rule-based Relation Construction from AMR.** We design a set of three rule-based patterns to extract canonical, fine-grained relations from the predicate-argument structure of AMR graphs. This provides a robust foundation for aligning question semantics with the KG structure.
- **AMR-aware Graph Structure Enhancement.** We introduce a method to inject the semantic links implicit in the AMR graph directly into the subgraph’s adjacency matrix. This creates ‘semantic shortcut’ for the GNN reasoner, effectively enhancing the efficiency and accuracy of multi-hop reasoning.
- **Transparent and Interpretable Reasoning Paths.** Our method significantly improves model interpretability by generating a clear reasoning chain from the topic entity to the answer entity. This path directly corresponds to the semantic path in the AMR graph, making the model’s decision process transparent and verifiable, rather than a ‘black-box’.

2. Related work

2.1. Subgraph retrieval for KBQA

Subgraph retrieval is a critical yet challenging step in modern KBQA, as directly reasoning over the entire KG is often computationally infeasible. Mainstream approaches aim to extract a small, relevant subgraph for the downstream reasoner. Early methods like GraftNet Sun et al. (2018) relied on heuristics, while subsequent neural models such as PullNet (Sun et al., 2019) and SR (Zhang et al., 2022) introduced trainable retrievers to better capture question semantics. However, a common limitation persists: these methods predominantly rely on shallow, embedding-based similarity, often failing to grasp the question’s complex semantic structure. This results in noisy subgraphs laden with irrelevant entities, which impairs reasoning performance and interpretability.

2.2. Semantic parsing in KBQA

To bridge this semantic gap, some research has turned to AMR, which provides a structured, canonical representation of a sentence’s core meaning, stripping away syntactic variations. Prior works have leveraged AMR primarily for full semantic parsing. For instance, NSQA Kapanipathi et al. (2021) transform questions into AMR-based query graphs to generate complete logical forms for neuro-symbolic reasoners, while QDAMR Deng et al. (2022) use AMR to decompose complex questions into simpler, solvable sub-questions. More recently, the structured nature of AMR has also been leveraged for sophisticated data augmentation. AMRFACT Qiu et al. (2024), for example, utilizes AMR as a semantic scaffold to systematically inject factual errors into correct summaries, thereby generating high-quality negative samples to train factuality classifiers.

In stark contrast to these methods, our work introduces a novel paradigm. Instead of using AMR to generate a query from scratch, we uniquely employ it as a high-precision semantic filter. We leverage the fine-grained relations extracted from the AMR graph to prune and refine an initially retrieved, potentially noisy subgraph. This approach directly targets the quality of the subgraph itself, enhancing reasoning accuracy at its root rather than compensating for noise in later stages.

2.3. GNNs for reasoning

The reasoning module in our framework is built upon GNNs, which exist within a broader context of Knowledge-Augmented Deep Learning (KADL). Standard deep learning models are often criticized for being data-hungry and lacking interpretability (Cui et al., 2025). KADL seeks to alleviate these deficiencies by integrating various forms of prior knowledge into deep models. Our approach aligns directly with this paradigm: we treat the linguistic structure captured by AMR as explicit semantic knowledge and integrate it (via our relation construction rules) to guide the neural GNN reasoner, enhancing both precision and interpretability.

More specifically, our method is an instance of Neuro-Symbolic AI, an increasingly active field focused on combining symbolic reasoning methods with deep learning to leverage their complementary benefits. Research in reasoning over KGs, which traditionally split between rule-based inference and embedding-based methods, is increasingly moving toward this neuro-symbolic paradigm (DeLong et al., 2025). Our framework directly contributes to this area by bridging the symbolic parse of a question (AMR) with a neural reasoner (GNN).

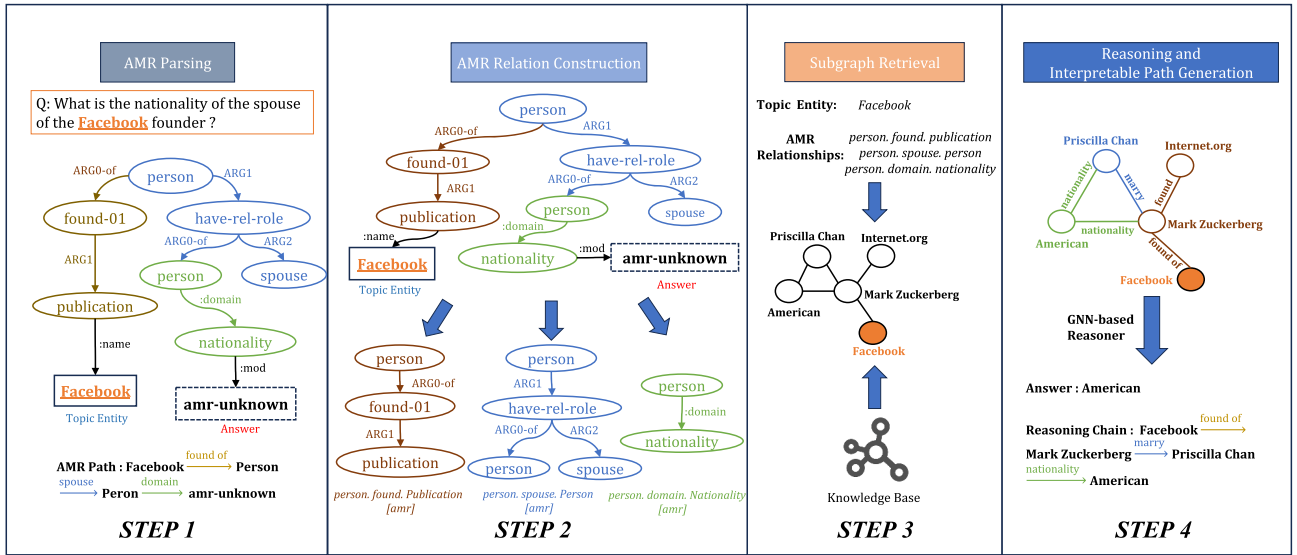


Fig. 2. An overview of our KBQA framework. Firstly, the question Q: “What is the nationality of the spouse of the Facebook founder?” is parsed using an AMR parser, converting it into an AMR graph. The path from the topic entity to the *amr-unknown* node is then explored, and the AMR relations are constructed based on the predicate nodes and semantic edges encountered along the path. Subsequently, semantic matching is performed with the relations in the knowledge graph to select the corresponding subgraph. Finally, the adjacency matrix of the subgraph is optimized using the AMR graph structure, and an AMR-based GNN reasoner is employed to predict the answer, replacing the corresponding nodes along the AMR path to obtain the reasoning chain.

Within this neuro-symbolic context, GNNs have emerged as a powerful tool for encoding the graph-structured data inherent to KGs and their subgraphs. Their ability to model complex relational information has made them a standard component in the subgraph QA ecosystem. For instance, recent work in the related task of question generation (QG) has successfully used GNNs to encode KG subgraphs, demonstrating their effectiveness in capturing structural information (Chen et al., 2023). This use of GNNs to benefit the downstream QA task affirms our choice of a GNN-based model (e.g., NSM or NuTrea) as the optimal neural reasoner to operate on the semantically-pure subgraphs our framework provides.

2.4. KG-enhanced LLM

LLMs have demonstrated remarkable capabilities across a wide range of complex tasks, including knowledge base question answering. Approaches such as ToG (Li et al., 2024), RoG (Pan et al., 2024), and ChatKBQA (Luo et al., 2023) leverage LLMs to enhance reasoning by decomposing questions or generating reasoning chains. However, these methods face inherent limitations: they often operate as black boxes with limited interpretability, and may suffer from hallucination issues when required knowledge is not well-represented in their training data.

To address these challenges, recent research has explored integrating KGs as external knowledge sources to improve LLM performance in KBQA tasks. KGs provide structured, factual representations that can enhance interpretability and reduce hallucination risks. Retrieval-augmented generation (RAG) methods (Lewis et al., 2020) represent one direction, typically focusing on textual or semi-structured knowledge sources. More recently, graph-structured approaches such as Graph-of-thought (Besta et al., 2024) and evidence pattern methods like EPR (Zhang et al., 2024) have shown that incorporating KG structure can significantly improve LLM reasoning.

AMR offers a promising approach as a structured semantic representation that can serve as auxiliary input for LLMs. By providing explicit semantic structures of questions, AMR can guide LLMs to perform step-by-step reasoning rather than direct answer seeking, reducing reasoning ambiguity and improving factual consistency. Our work demonstrates how AMR can be integrated with LLMs to enhance KBQA performance, as discussed in Section 7.

3. Method

Fig. 2 illustrates the pipeline of our AMR-based KBQA system, which consists of four modules: *i*) AMR Parsing, which transforms question text into AMR graphs to obtain structured semantic information. *ii*) AMR Relation Construction, which leverages the semantic structure of the AMR graph to extract the relations needed to answer the question. *iii*) Subgraph Retrieval, which performs semantic matching between extracted relations and relations in the KG to retrieve the relevant subgraph. *iv*) Reasoning and Interpretable Path Generation, which performs reasoning over the refined subgraph to predict the final answer, and then constructs an interpretable reasoning path guided by the AMR’s structure.

3.1. AMR parsing

An AMR graph is a rooted, directed, and acyclical structure. The nodes in an AMR graph represent different concepts, i.e., named entities, quantities, dates, and other phenomena (Banarescu et al., 2013); Edges represent relations between these concepts, such as :domain and :name (Kapanipathi et al., 2021). An AMR path can be obtained from the topic entity to the answer entity, where the nodes and edges along this path capture the semantic information necessary to answer the question. This semantic information is then used for the construction of AMR relations. The following is the path we extract based on AMR parsing:

Facebook $\xrightarrow{\text{found of}}$ Person $\xrightarrow{\text{spouse}}$ Person $\xrightarrow{\text{domain}}$ unknown

We leverage AMR parsing to capture the semantic structure of questions. Fig. 2 illustrates the AMR graph for the question: “What is the nationality of the spouse of the Facebook founder?”. *amr-unknown* and *Facebook* are two key entities involved in answering the question, where *amr-unknown* represents the unknown answer entity and *Facebook* serves as the topic entity extracted from the question. These two entities correspond to the starting point and final target in the KBQA reasoning process.

For AMR parsing, we adopt SPRING (Bevilacqua et al., 2021), a widely used method for AMR parsing. A key advantage of SPRING is its seamless integration with BLINK (Wu et al., 2020), a highly effective entity linking tool. BLINK directly annotates topic entities in the AMR graph with their KG counterparts, eliminating the need for separate entity linking.

3.2. AMR relation construction

A high-quality subgraph must contain a path from the topic entity to the correct answer, while minimizing noisy paths that could mislead the reasoner. Our approach leverages the rich structure of the question's AMR graph to filter the subgraph. The core of this process is identifying predicates and their arguments from the AMR to form canonical relations. We developed an algorithm (Algorithm 1) to extract these relations by analyzing the shortest path from the topic entity to the *amr-unknown* node. Crucially, the relation paths derived from the AMR graph serve as the core semantic constraints for downstream filtering. Therefore, when multiple shortest paths of equal topological length exist in the AMR graph, the predicate-argument structures along all such paths are considered to possess equivalent semantic utility for identifying the correct answer. To prevent premature information loss and maximize the recall of target relations, our algorithm comprehensively retains the relations from all qualifying shortest paths, merging them into a unified candidate set via a union operation before entering the downstream matching phase. This process identifies three primary patterns for relation construction, as illustrated in Fig. 2. The following sections will provide a detailed explanation of these three construction patterns.

3.2.1. Predicate-centric

This pattern targets the most common linguistic structure where an action or state, represented by a predicate node (e.g., *found-01*), links two or more participants (the brown part in Step 2 of Fig. 2). According to PropBank framesets (Kingsbury & Palmer, 2002), the adjacent attribute edges like *ARGx* define the semantic roles of these participants. We treat the entity connected by the edge with a smaller parameter label as the agent of the action, and the other as the recipient. This allows us to extract a directed, typed relation; for example, the *found-01* predicate in our running example robustly yields the relation *person.found.publication*.

3.2.2. Role-centric

Natural language often expresses relationships through nominal roles rather than direct actions. To handle cases like “spouse of the founder”, which are represented in AMR by a predicate (e.g., *have-rel-role-91*) with an auxiliary attribute (e.g., *spouse* connected via *ARG2*), this pattern is designed (the blue part in Step 2 of Fig. 2). It simplifies the structure by replacing the intermediate predicate node with the more meaningful auxiliary attribute itself. This effectively transforms the structure into the first predicate-centric pattern, allowing for a consistent relation extraction mechanism.

3.2.3. Edge-centric

Some semantic relationships are directly encoded as edges in the AMR graph, rather than through predicate nodes. This pattern addresses relations like time, location, or possession (e.g., the *:domain* edge), as shown in the green part in Step 2 of Fig. 2. When a guiding predicate is absent, we determine the relation's direction by leveraging the natural path flow from the topic entity towards the *amr-unknown* node, ensuring the extracted relation aligns with the question's logical intent.

Generality of Rule-based relations. The three rule-based patterns we propose (predicate-centric, role-centric, and edge-centric) are designed to capture the most common semantic structures found in natural language questions. These patterns are derived from a comprehensive analysis of PropBank frameset annotations (Kingsbury & Palmer, 2002), which

cover the majority of verb argument structures in English. The rule-based approach offers several advantages: it provides predictable behavior, requires minimal training data, and enables easy inspection and debugging.

To assess the coverage of our rules, we analyzed the AMR graphs of all questions in the WebQSP and CWQ datasets. Our three patterns successfully cover **96.2%** of the semantic structures needed to answer these questions. The remaining cases (3.8%) typically involve complex compositional structures, domain-specific predicates, or linguistic phenomena not fully covered by the standard PropBank frameset.

Algorithm 1 Relation construction based on AMR.

```

1: Input: Question text  $Q$ 
2: Output: AMR relation set  $R$ 
3:  $G \leftarrow \text{AMR Parsing}(Q)$ 
4:  $E \leftarrow \text{Entity Linking}(G)$ 
5: if  $E \neq \emptyset$  and  $\text{amr-unknown} \in G$  then
6:   for each  $e$  in  $E$  do
7:      $P_e \leftarrow \text{getShortestPath}(G, e, \text{amr-unknown})$ 
8:   end for
9:   for each  $v$  in  $V_p$  do
10:    if  $\text{isVerb}(v)$  then
11:       $(e_1, e_2) \leftarrow \text{getNeighborNodes}(v)$ 
12:      buildRelation $(e_1, e_2, v)$ 
13:    else if  $\text{isRoleVerb}(v)$  then
14:       $(e_1, e_2) \leftarrow \text{getNeighborNodes}(v)$ 
15:       $e_3 \leftarrow \text{getRole}(v)$ 
16:      buildRelation $(e_1, e_2, e_3)$ 
17:    end if
18:   end for
19:   for each  $r$  in  $R_p$  do
20:    if  $\text{isRelation}(r)$  then
21:       $(e_1, e_2) \leftarrow \text{getEntity}(r)$ 
22:      buildRelation $(e_1, e_2, r)$ 
23:    end if
24:   end for
25: end if
26: return  $R$ 

```

The primary variables and functions used in Algorithm 1 are defined as follows:

- Q : The input natural language question text.
- R : The final output set of AMR-based relations constructed by the algorithm.
- G : The AMR graph generated by parsing the input question Q .
- E : The set of topic entities identified in the graph G through entity linking.
- e : A single topic entity from the set E .
- P_e : The shortest path extracted from the AMR graph G , running from a topic entity e to the *amr-unknown* node.
- V_p : The set of predicate nodes (verbs or role-verbs) found along the extracted path(s) P .
- v : A single predicate node from the set V_p .
- R_p : The set of relation edges (e.g., semantic edges like *:domain*) found along the extracted path(s) P .
- r : A single relation edge from the set R_p .
- e_i : Variables representing entities or concepts that are arguments to a predicate, used to construct the final relations in R .

The relation construction may also involve the *amr-unknown* node, which is substituted with the expected answer type (e.g., *person*, *place*) based on the question's starting word (e.g., ‘who’, ‘where’).

3.3. Subgraph retrieval

We use the fine-grained relations extracted from the AMR graph to guide the subgraph refinement process. Originating from a deep semantic representation of the question, these AMR relations act as a versatile, high-precision filter. This allows our method to be flexibly applied to any initially retrieved (and potentially noisy) subgraph to prune it down to the most semantically relevant entities and relations. For any subgraph containing several triples $\mathcal{G} = \{(h, r, t) \mid h, t \in \mathcal{E}, r \in \mathcal{R}\}$, where $\mathcal{E}$ and $\mathcal{R}$ denote the entity set and the relation set, respectively. Let $\mathcal{E}_r = \{e_1, e_2, \dots, e_n\}$, which denotes the n entity types available in AMR, and let $\mathcal{R}_i = \{r_1^i, r_2^i, \dots, r_m^i\}$, which denotes the set of relations where e_i serves as the head entity. For each entity $e \in \mathcal{E}$, we check if its type is in $\mathcal{E}_r$. If the corresponding entity type e_i is present, we retain the entity and then determine whether the entity's relations should be preserved; otherwise, the entity is excluded.

Entity Type Alignment and Robustness. The entity types extracted from AMR (e.g., 'person', 'place') serve as semantic filters for subgraph retrieval. While the ontological schema of Freebase (the underlying KB for our datasets) and AMR are not directly mappable at a fine-grained level, our method employs a hierarchical type generalization strategy to ensure robust alignment.

Specifically, we implement a type compatibility function that checks both exact matches and hierarchical parent-child relationships. For instance, an entity typed as 'film.director' in Freebase is compatible with the AMR type 'person' through the hierarchical mapping. To address potential misalignments that could lead to filtering out critical triples, we adopt a conservative pruning strategy: entities that are directly connected to the topic entity are preserved regardless of type mismatch, while more distant entities are subject to type-based filtering. This ensures that the answer coverage remains high (92.13% for PPR + AMR, where PPR represents the Personalized PageRank algorithm (Haveliwala, 2003), and 90.30% for SR + AMR, where SR denotes the embedding-based Subgraph Retrieval retriever (Zhang et al., 2022), as shown in Table 2) while still effectively removing noisy entities.

Additionally, we maintain a set of common type mapping patterns derived from PropBank schema annotations to bridge the gap between AMR predicates and Freebase relation types. This hybrid approach—combining structural alignment from AMR with flexible type matching—enables our method to effectively filter noise while preserving essential entities for answering the question.

For each relation $r_i \in \mathcal{R}_i$, we compute its correlation with the relation of the retained entity e (denoted as r'), obtain its embedding using the pre-trained model BERT (Kenton & Toutanova, 2019), and measure the correlation using cosine similarity:

$$\text{Similarity}(r, r') = \frac{h(r) \cdot h(r')}{|h(r)| |h(r')|} \quad (1)$$

where $h(\cdot)$ denotes the embedding function instantiated by BERT, and the hyperparameter δ is the threshold for subgraph retrieval. The above operations for each entity will result in a subgraph that includes topic entities, type-specific entities, and question-related relations.

While our framework is fundamentally structure-driven, we incorporate embedding similarity as a pragmatic mechanism for matching AMR-derived relations with KG relations that may not have identical textual labels. The similarity threshold δ serves as a flexible matching criterion that accounts for lexical variations and synonymous expressions. For instance, the AMR predicate 'found-01' may map to multiple KG relations like 'people.founder', 'organization.founder', or 'people.founded', and embedding similarity enables robust matching across these variations.

3.4. Reasoning over subgraphs

After retrieval of the subgraph using our AMR-based method, we introduce an AMR-informed structural enhancement to incorporate the

explicit semantic structure of AMR graphs, thereby improving reasoning efficiency.

GNN models are well-suited for handling multi-hop reasoning in KGs, where entities must be contextually linked. However, standard KGs may lack explicit links between semantically related entities, which may cause longer reasoning chains and less efficient reasoning. Since AMR provides a structured semantic representation, it can reveal implicit relations that are not directly encoded in the KG. To integrate this additional information, we modify the adjacency matrix of the retrieved subgraph $\mathcal{G}_Q = (\mathcal{E}_Q, \mathcal{R}_Q)$ as follows:

1. **Identifying AMR-Linked Nodes:** Given an AMR graph $\mathcal{A}_Q$ derived from the question, we first identify pairs of nodes (e_i, e_j) that are connected in $\mathcal{A}_Q$ but not in $\mathcal{G}_Q$.
2. **Graph Structure Modification:** If e_i and e_j are not adjacent in $\mathcal{G}_Q$, we add an edge (e_i, e_j) to the adjacency matrix.
3. **Semantic Edge Labeling:** The newly added edge is assigned with a special relation label [amr], ensuring its differentiation from standard KG relations and enabling the model to recognize AMR-derived semantic dependencies.

Formally, we update the adjacency matrix A of the subgraph:

$$A'_{i,j} = \begin{cases} 1, & \text{if } (e_i, e_j) \in \mathcal{G}_Q \text{ or } \in \mathcal{A}_Q, \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

Similarly, we modify the relation matrix R by setting:

$$R'_{i,j} = \begin{cases} [\text{amr}], & \text{if } (e_i, e_j) \in \mathcal{A}_Q \text{ and } \notin \mathcal{G}_Q, \\ R_{i,j}, & \text{otherwise.} \end{cases} \quad (3)$$

During message passing, the feature update equation follows the standard GNN formulation but now includes the newly introduced [amr] edges:

$$\mathbf{h}_e^{(l)} = \sigma \left(\sum_{r \in \mathcal{R}_Q \cup \{[\text{amr}]\}} \sum_{e' \in \mathcal{N}_r(e)} W_r^{(l)} \mathbf{h}_{e'}^{(l-1)} + W_0^{(l)} \mathbf{h}_e^{(l-1)} \right) \quad (4)$$

where $\mathcal{N}_r(e)$ now includes both standard KG relations and AMR-derived relations, and $W_r^{(l)}$ represents the transformation matrix for the type of relation r at layer l .

This modification enhances the capability of the GNN model to propagate information efficiently by explicitly integrating AMR-informed semantic structures. The updated adjacency matrix allows the model to directly attend to semantically related entities that would otherwise require multi-hop reasoning.

3.5. Interpretable path generation

Unlike traditional methods that only produce a final answer entity, our framework leverages the AMR structure to generate a interpretable reasoning path. The core idea is to instantiate the abstract AMR path (e.g., 'Facebook' $\rightarrow$ 'Person' $\rightarrow$ 'Person' $\rightarrow$ 'unknown') with concrete entities from the retrieved subgraph.

A naive approach to this instantiation would be to simply find the shortest path in the subgraph connecting the topic and answer entities. However, this often fails as the topologically shortest path may not be semantically correct. For example, given the question: "Where does the spouse of the founder of Facebook live?", a shortest-path approach might yield an incorrect chain. An ideal reasoning path, in contrast, should be:

Facebook $\rightarrow$ founder $\rightarrow$ spouse $\rightarrow$ residence

However, due to structural constraints in the KG, the shortest path might be:

Facebook $\rightarrow$ founder $\rightarrow$ residence

Table 1
Statistics of the CWQ and WebQSP dataset.

Dataset	#Train	#Val.	#Test
CWQ	27,639	3519	3531
WebQSP	2848	250	1639

which incorrectly skips the spouse relation, leading to an invalid reasoning chain. To address this issue, we propose a heuristic search constrained by entity types to construct the reasoning paths.

Rather than enforcing strict relation consistency between AMR and the KG, we prioritize entity type consistency when selecting the reasoning path. Given an AMR-derived entity sequence $T_A = \{e_1, e_2, \dots, e_m\}$, and a candidate reasoning path in the retrieved subgraph $P = \{p_1, p_2, \dots, p_n\}$, we define a path score $S(P)$ based on the number of matched entity types:

$$S(P) = \frac{|T_A \cap P|}{|T_A|} \quad (5)$$

where $|T_A \cap P|$ represents the number of matched entity types. A higher $S(P)$ indicates greater alignment between the path and the entity types in the ideal reasoning path.

We employ a heuristic search strategy for efficient determination of the optimal reasoning path. First, we extract multiple candidate paths from the topic entity to the answer entity in the retrieved subgraph. Then, we compute the entity-type-based score $S(P)$ by evaluating each path's alignment with the AMR-derived entity sequence. Finally, we select the path with the highest score. If there are multiple paths sharing the highest score, we choose the shortest one to ensure minimal reasoning steps.

After obtaining the optimal path, similar to entity linking, the abstract entity types in the AMR graph are sequentially linked to specific entities on the path, resulting in an interpretable reasoning chain.

Finally, we generate the reasoning path by replacing nodes and edges we obtain in the AMR parsing step, as shown in Step 4 of Fig. 2:

Facebook $\xrightarrow{\text{found of}}$ Mark Zuckerberg $\xrightarrow{\text{marry}}$
Priscilla Chan $\xrightarrow{\text{nationality}}$ American

4. Experiments

In this section, we designed experiments to answer the following key questions: (1) Can AMR-extracted relations contribute to higher-quality subgraph retrieval? (2) To what extent does AMR-based semantic extraction improve QA performance? (3) How effective is graph structure optimization in enhancing the reasoning accuracy?

4.1. Datasets

We evaluated our method on two widely used benchmarks: WebQuestionsSP (WebQSP) (Yih et al., 2015) and Complex WebQuestions 1.1 (CWQ) (Talmor & Berant, 2018). Both datasets are built on the Freebase knowledge base (Bollacker et al., 2008). The former mostly contains simple questions, while the latter includes more complex questions. Table 1 presents the statistics.

Table 2
Answer coverage and subgraph size comparison.

Method	Answer Coverage	Nodes	Relations
AMR	92.37%	183,423	7983
PPR	94.87%	1,441,420	6102
PPR + AMR	92.13%	581,840	3222
SR	91.22%	183,433	26,282
SR + AMR	90.30%	108,362	5630

4.2. Metric

For the KBQA task, an ideal subgraph should include paths from the topic entities to the target answers, with minimal noisy paths that may lead to incorrect entities. Since the ground truth of the path is often difficult to obtain, we evaluated the subgraph quality using two metrics: the size of the retrieved subgraph and the answer coverage rate, which represents the proportion of questions for which the subgraph contains at least one answer entity.

For the reasoning performance, we follow the practice of recent studies (Sun et al., 2018), which consider reasoning as a ranking task. For each test question, we ranked all candidate entities based on the prediction probability of the evaluated model, and then determined whether the top-1 answer is correct using Hits@1. Considering that a question may have multiple answers, we also adopted the commonly used F1 metric.

4.3. Implementation details

We implement our model using the PyTorch framework. For AMR parsing, we utilize SPRING (Bevilacqua et al., 2021) integrated with the BLINK entity linker (Wu et al., 2020). The similarity threshold δ for our AMR-based retriever was tuned on the development set and set to 0.67 for WebQSP and 0.64 for CWQ. For all GNN-based reasoner, we adopted the hyper-parameter settings from their original papers to ensure a fair comparison.

4.4. Baselines

To comprehensively evaluate our framework, we compare it against a set of representative and state-of-the-art KBQA models. We provide a detailed description of each model used in our comparison below:

GraftNet (Sun et al., 2018). uses a variation of GCN to update the entity embedding and predict the answer.

PullNet (Sun et al., 2019). improves GraftNet by retrieving relevant documents and extracting more entities to expand the entity graph.

EmbedKGQA. (Saxena et al., 2020) incorporates pre-trained knowledge embedding to predict answer entities based on the topic entity and the query question.

NSM. (He et al., 2021) makes use of the teacher framework to provide the distribution of intermediate entities as supervision signs for the student network. However, NSM fails to utilize the semantic connection between relations and information from head-tail entities to enhance the construction of current relation representation.

UniKGQA. (Jiang et al., 2023) unifies the retrieval and reasoning in both model architecture and learning for the multi-hop KGQA.

RAH-KBQA. (Du et al., 2023) proposes a framework for unified reasoning of a primal entity graph and a dual relation graph. It enhances relation representation by incorporating semantic information from head-tail entities and the co-relations among relations. The model iteratively performs reasoning on the primal entity graph, propagates information on the dual relation graph, and facilitates interaction between the two to refine entity and relation representations.

NuTrea. (Choi et al., 2023) introduces a tree search-based GNN model that incorporates broader knowledge graph context. It uses a message-passing scheme that considers future-oriented subtree information to enrich past-oriented embeddings during path searching. Additionally, it proposes the Relation Frequency-Inverse Entity Frequency (RF-IEF) node embedding to better characterize ambiguous nodes using global KG statistics.

4.5. Subgraph quality evaluation

We compared three types of subgraphs: (a) Subgraphs retrieved by our method, denoted as AMR, which extracts fine-grained relations to filter knowledge graph paths; (b) Subgraphs obtained by baseline methods, which serve as the initial, general-purpose retrievers in our experiments, including: 1) PPR model: personalized PageRank algorithm (Haveliwala, 2003), and 2) SR model: embedding-based subgraph retrieval retriever (Zhang et al., 2022); and (c) Subgraphs obtained by baseline methods and refined using AMR constraints, designated as PPR + AMR and SR + AMR, which represent the semantically filtered and pruned compact subgraphs, respectively.

The results in Table 2 clearly demonstrate the effectiveness of our AMR-based filtering. When comparing SR + AMR directly to the baseline SR, our method achieves a remarkable reduction in subgraph size, decreasing the number of nodes by 40.9% and relations by 78.6%. This indicates that our AMR filter successfully prunes a massive amount of irrelevant noise. Crucially, this aggressive pruning only results in a minimal 0.92% drop in Answer Coverage, confirming that our AMR-derived relations preserve the essential components needed for the answer while discarding noise. While PPR-based methods achieve higher initial coverage, they do so at the cost of enormous subgraph sizes (1.44M nodes). Our AMR refinement (PPR + AMR) also proves effective here, cutting the node count by nearly 60% and the relation count by almost 50%.

Impact of Subgraph Quality on QA Performance. To directly demonstrate how our AMR-based refinement improves final QA performance, we conducted an ablation study comparing different subgraph retrieval strategies with the same reasoner. As shown in Table 3, using our AMR-refined subgraphs consistently yields higher QA accuracy compared to using raw subgraphs from baseline retrievers.

For example, with the NSM reasoner, our AMR-retrieved subgraphs improve Hits@1 by **3.1 points** on CWQ and **6.7 points** on WebQSP compared to the SR retriever. This confirms that higher subgraph quality (measured by compactness and answer relevance) directly translates to better reasoning performance.

Table 3

End-to-end QA performance with different subgraph retrieval strategies using the NSM reasoner.

Retrieval Method	CWQ		WebQSP	
	Hits@1	F1	Hits@1	F1
SR Baseline	49.3	46.3	69.5	64.1
SR + AMR	52.4	51.8	76.2	71.1
PPR Baseline	47.2	43.8	67.8	62.4
PPR + AMR	50.6	51.2	75.7	73.7

Table 4

Performance comparison on the CWQ and WebQSP datasets. Hits@1 and F1 scores are reported. The best overall results are in **bold**. Results for our models are underlined.

Method	CWQ		WebQSP	
	Hits@1	F1	Hits@1	F1
KV-Mem	18.4	15.7	46.6	34.5
EmbedKGQA	32.0	–	66.6	–
PullNet	45.9	–	68.1	–
GraftNet	36.8	32.7	66.4	60.4
NSM	47.6	42.4	68.5	62.8
SR + GCN	49.1	42.7	66.7	63.1
SR + NSM	49.3	46.3	69.5	64.1
UniKGQA + NSM	49.2	–	69.1	–
UniKGQA + UniKGQA	50.7	48.0	75.1	70.2
EPR + NSM	60.6	61.2	71.2	70.2
NuTrea	53.6	49.5	77.4	72.7
RAH-KBQA	54.4	49.3	77.2	72.6
Our Models				
AMR + NuTrea*	<u>55.6</u>	<u>52.4</u>	78.1	73.8
AMR + NSM*	<u>54.8</u>	<u>52.1</u>	<u>73.6</u>	<u>72.5</u>

The improvement is particularly pronounced on multi-hop questions, where noisy paths in the subgraph can easily mislead the reasoner.

Furthermore, comparing PPR-based methods, our AMR refinement improves Hits@1 by **3.4 points** on CWQ and **7.9 points** on WebQSP, demonstrating that the benefits of our approach are consistent across different initial retrieval strategies. This validates that our AMR-driven filtering effectively removes noise that would otherwise degrade the downstream reasoning performance.

4.6. QA performance evaluation

We evaluate our framework by pairing our AMR-based retriever with existing subgraph-oriented reasoners. Models using our AMR-informed structural enhancement are denoted with an asterisk (*); for example, NSM* is our modified version of the NSM reasoner.

As presented in Table 4, our framework demonstrates exceptional performance, establishing a new state-of-the-art on the widely-used WebQSP benchmark. Specifically, our AMR + NuTrea* model achieves a Hits@1 score of 78.1 and an F1 score of 73.8, surpassing all prior methods including strong recent baselines like NuTrea and RAH-KBQA. This result underscores the significant benefits of our AMR-based subgraph refinement. Furthermore, our AMR + NSM* model also shows substantial gains, improving upon the SR + NSM baseline by 4.1 points in Hits@1 and 8.4 points in F1 score.

On the more complex CWQ dataset, our models remain highly competitive. Our AMR + NuTrea* model outperforms other retrieval-based GNN baselines like NuTrea and RAH-KBQA. This confirms that our AMR-based refinement effectively

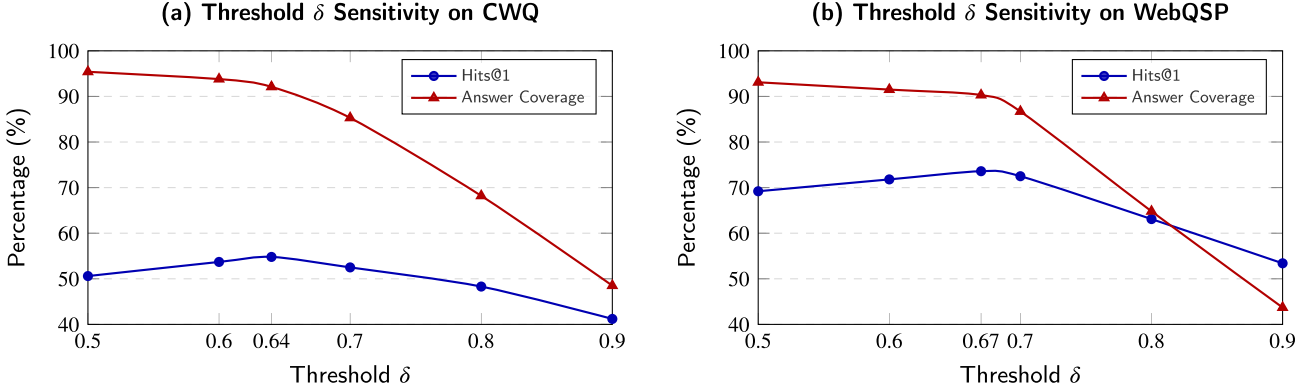


Fig. 3. Sensitivity analysis of the relation matching threshold δ on the CWQ and WebQSP datasets using the AMR + NSM* framework.

Table 5

Impact of adjacency matrix modification on QA performance across different retrieval methods.

Method	CWQ		WebQSP	
	H@1	F1	H@1	F1
SR + NSM	49.3	46.3	69.5	64.1
SR + NSM*	50.4	47.1	70.2	64.8
AMR + NSM	51.7	49.8	71.2	67.2
AMR + NSM*	53.8	52.1	72.6	71.3
EPR + NSM	60.6	61.2	71.2	70.2
EPR + NSM*	61.2	61.8	72.3	71.8

Table 6

Ablation analysis evaluated exclusively on the specific subsets triggered by adjacency matrix modification (using the NSM reasoner).

Method (Triggered Subset)	CWQ Subset (48.7%)		WebQSP Subset (29.4%)	
	Hits@1	F1	Hits@1	F1
AMR + NSM (w/o [amr] edge)	49.5	47.2	70.1	66.4
AMR + NSM* (w/ [amr] edge)	53.8	51.8	72.9	69.8

enhances subgraph quality even for challenging multi-hop questions. Notably, our approach, along with EPR, exhibits superior performance compared to other methods. While EPR's KG-centric evidence patterns give it an edge on CWQ, both methods are founded on the same core intuition: transforming unstructured text into fine-grained semantic structures is key to robust reasoning. The success of both paradigms validates our question-centric AMR approach, suggesting that structured subgraph construction is highly effective at eliminating noise.

4.7. Sensitivity analysis of similarity threshold

To systematically investigate the impact of the relation matching threshold δ (defined in Eq. 1), we conducted a sensitivity analysis on both the CWQ and WebQSP datasets by varying δ from 0.50 to 0.90. The choice of δ governs a delicate trade-off between subgraph compactness and semantic recall.

As illustrated in Fig. 3, when δ is configured too low (e.g., 0.50), the retriever tolerates excessive lexical variations, introducing severe relational noise into the retrieved subgraph and consequently dragging down the reasoning performance (Hits@1) across both benchmarks. Conversely, when δ is set excessively high (e.g., ≥ 0.80), the filtering criteria become overly restrictive, causing a steep decline in Answer Coverage as valid semantic paths are inadvertently pruned, which in turn leads to a drop in Hits@1. The optimal trade-off is achieved at $\delta = 0.64$ for CWQ and $\delta = 0.67$ for WebQSP, balancing subgraph purity with factual recall.

4.8. Effect of adjacency matrix modification

Table 5 demonstrates the impact of our AMR-informed adjacency matrix modification on QA performance. This structural enhancement consistently improves reasoning accuracy across

all tested retrieval methods. For the SR-based retriever, applying our modification (SR + NSM*) improves the H@1 and F1 scores by 1.1 and 0.8 points on CWQ, and by 0.7 points for both metrics on WebQSP, respectively.

The performance gains are even more pronounced for our own AMR-based retriever. Specifically, our AMR + NSM* model surpasses its un-enhanced counterpart (AMR + NSM) by a significant margin, improving H@1 by 2.1 points and F1 by 2.3 points on CWQ. The effect is particularly notable on WebQSP, where the F1 score increases by 4.1 points. These results suggest that the enhancement informed by AMR is significantly more effective in improving final answer accuracy.

This trend also holds for the powerful EPR retriever, confirming the general applicability of our method. Overall, these results validate that incorporating AMR-derived connections into the graph structure effectively captures implicit semantic relations, thereby improving both the precision and recall in KBQA.

Statistically, the AMR-informed adjacency matrix modification mechanism was triggered in 48.7% of the samples in the complex CWQ dataset, whereas it was triggered in 29.4% of the samples in the simpler WebQSP dataset. This discrepancy is highly aligned with our intuition that complex, multi-hop questions inherently encapsulate richer structural variations and implicit semantic constraints that require structural shortcuts.

To rigorously isolate and evaluate the explicit efficacy of introducing the [amr] edges, we conducted a targeted ablation study restricted exclusively to these triggered subsets (i.e., the portion of data that actually underwent adjacency matrix modifications). As presented in Table 6, when evaluated solely on the CWQ triggered subset, the AMR + NSM* model outpaces its un-enhanced counterpart (AMR + NSM) by 4.3 points in Hits@1 and 4.6 points in F1 score. This performance gain is substantially more pronounced than the overall margin observed on the entire

Table 7

Impact of AMR-augmented relations on LLM-based KBQA methods on CWQ dataset. “1-hop” uses AMR relations directly connected to topic entity (single-hop constraints); “2-hop” extends to two-hop relations; “all” uses all AMR-constructed relations.

AMR Augmentation	RoG		ToG	
	Hits@1	F1	Hits@1	F1
Original	62.6	57.1	60.3	55.8
1-hop	64.8	59.2	62.5	58.1
2-hop	65.2	60.4	64.0	60.7
all	65.7	61.2	65.1	62.4

dataset. This empirical evaluation solidifies the conclusion that integrating structural constraints directly into the GNN’s topological message-passing space drastically mitigates the difficulty of multi-hop relational reasoning.

4.9. Comparison with LLM-based methods

Recent research has explored integrating knowledge graphs with LLMs to enhance KBQA performance through retrieval-augmented mechanisms. To validate the effectiveness of AMR in constraining reasoning, we integrate the relations constructed by our AMR-driven approach into existing LLM-based KBQA methods as auxiliary inputs.

We conduct experiments on the CWQ dataset, which contains a substantial number of multi-hop questions, making it suitable for evaluating the impact of structured semantic constraints. As shown in Table 7, injecting AMR relations significantly improves model accuracy across different LLM methods.

The table compares three augmentation strategies: (1) ‘1-hop’ uses only AMR-constructed relations that are directly connected to the topic entity, representing single-hop semantic constraints; (2) ‘2-hop’ extends this to two-hop relations, providing multi-hop constraints; and (3) ‘all’ incorporates all AMR-constructed relations. The ‘Original’ row represents the baseline without AMR augmentation.

The results demonstrate that AMR augmentation consistently improves Hits@1 performance for both RoG and ToG methods. Notably, the ‘1-hop’ strategy achieves the most significant gains, with RoG improving from 62.6 to 64.8 (+ 2.2) and ToG improving from 60.3 to 62.5 (+ 2.2). This indicates that even a minimal set of directly relevant AMR relations provides strong guidance that helps models identify correct answers. The ‘2-hop’ and ‘all’ strategies show additional but diminishing improvements, suggesting that including more relations offers limited additional benefits once core semantic constraints are established.

These findings confirm that AMR-constructed relations provide question-relevant constraints that complement the textual knowledge typically provided to LLMs. Compared to naive text-based KG inputs, our AMR-derived relations offer more precise semantic alignment with the question structure, helping models navigate the reasoning space more effectively and reducing ambiguity in multi-hop scenarios.

4.10. Inference time analysis

To evaluate the overall efficiency of our framework, we measured the end-to-end average inference time per question on the WebQSP test set. The timing includes all steps from subgraph retrieval to final answer generation. All methods were benchmarked on the same server equipped with a single NVIDIA Tesla

Table 8

End-to-end inference time comparison. *PullNet’s inference time could occasionally extend to 5–6 seconds on questions resulting in particularly dense subgraphs.

Method	Average Inference Time (s/question)
PullNet	2.0 - 3.0 s*
SR	1.0 s
Our Method	1.0 s

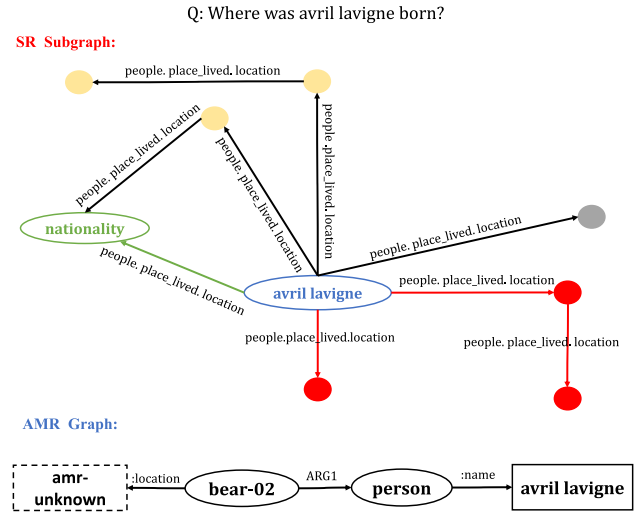


Fig. 4. Subgraph of the question “Where was Avril Lavigne born?” obtained based on SR and its corresponding AMR graphs.

V100 GPU (32GB memory). The results are summarized in Table 8.

As shown in the table, the average inference time of our method is 1.0 second per question, which is on par with the highly efficient SR retriever. This demonstrates that the enhanced subgraph quality provided by our AMR-driven approach does not introduce significant computational overhead. In contrast, methods like PullNet, which retrieve full 2-hop neighborhoods before applying computationally expensive pruning techniques, are notably slower, especially in worst-case scenarios. Our efficient performance highlights the practicality of our framework for real-world applications.

4.11. Case study

Our method’s superiority is evident in cases like the question “Where was Avril Lavigne born?”. As shown in Fig. 4, conventional retrievers like SR, relying on shallow textual similarity, misinterpret ‘born’ and retrieve numerous incorrect paths based on the *people.place.lived.location* relation, flooding the subgraph with irrelevant nodes. In contrast, our approach first parses the question into its core semantic AMR graph, which correctly represents ‘born’ with the *bear-02* predicate. From this, we extract the canonical relation *person.bear.place* to act as a precise filter. This filter prunes all semantically unaligned paths, leaving a clean subgraph with only the correct path to the answer, ‘Canada’. The final interpretable chain is simply:

Avril Lavigne ^{bear}→ Canada.

To further illustrate how our approach leverages AMR to understand complex semantic structures, we provide additional

examples of multi-hop questions and their corresponding AMR graph representations in PENMAN notation. The indentation in the graphs is significant as it represents the hierarchical semantic structure.

Q1:. In which city was the director of ‘The Dark Knight’ born?

```
(z1 / bear-02
  :ARG1 (z2 / person
    :ARG0-of (z3 / have-org-role-91
      :ARG1 (z4 / film
        :name (z5 / name
          :op1 "The"
          :op2 "Dark"
          :op3 "Knight")))
    :ARG2 (z6 / director)))
  :location (z7 / city
    :mod (z8 / amr-unknown)))
```

Q2:. What country does the director of ‘Pulp Fiction’ come from?

```
(z1 / come-01
  :ARG1 (z2 / person
    :ARG0-of (z3 / have-org-role-91
      :ARG1 (z4 / publication
        :name (z5 / name
          :op1 "Pulp"
          :op2 "Fiction")))
    :ARG2 (z6 / director)))
  :ARG3 (z7 / country
    :mod (z8 / amr-unknown)))
```

Q3:. In which country was ‘Inception’ filmed?

```
(z1 / film-01
  :ARG1 (z2 / film
    :name (z3 / name
      :op1 "Inception")))
  :location (z4 / country
    :mod (z5 / amr-unknown)))
```

This case highlights a fundamental difference from other methods. While approaches like RAH-KBQA and NuTrea focus on refining reasoning over a potentially noisy graph, our method tackles the problem at its root by improving the input graph’s quality *before* reasoning begins. By providing the reasoner with a semantically pure subgraph, we reduce the computational burden and significantly lower the risk of error propagation.

Despite our framework’s superiority, a small fraction of failure cases (3.8% of the dataset) occur due to the limitations of upstream AMR parsers. This error propagation in pipeline-based neuro-symbolic designs typically upper-bounds downstream reasoning precision through two representative bottlenecks:

(1) *Modifier Attachment Ambiguity*: In inverted multi-hop queries like “Of the two movies directed by Nolan in 2010, which one starred Leonardo?”, the parser may mislink the temporal modifier 2010 to the starring rather than the directing event, yielding an invalid relation sequence (*movie.starred.in.year*) and causing over-aggressive pruning. (2) *Lexical Polysemy and Frameset Confusion*: For queries with polysemous verbs (e.g., “Which country backed the project?”), the parser might misalign the predicate with a spatial orientation frameset (back-01) instead of a supportive one (back-02), failing to match knowledge graph schemas like *projects.supported.by*.

To mitigate these errors, two optimization strategies can be deployed. First, text-based pre-processing can simplify queries before parsing (e.g., converting inverted structures into active voice) to reduce structural ambiguity. Second, establishing a bidirectional feedback loop between the GNN reasoner and the parser can allow downstream consistency scores to dynamically rectify upstream parsing discrepancies, thereby enhancing the overall robustness.

5. Conclusion

In this paper, we addressed the critical semantic gap in retrieval-based KBQA, where conventional methods relying on text-embedding similarity often produce noisy subgraphs that impair reasoning. To overcome this, our work introduces a novel AMR-driven framework that leverages Abstract Meaning Representation as a powerful intermediate semantic scaffold. We proposed a rule-based method to construct canonical, fine-grained relations from AMR graphs, which serve as high-precision filters to prune initial subgraphs into more compact and semantically coherent versions for the reasoner. Furthermore, we demonstrated how the AMR graph’s structure can be directly incorporated into the subgraph’s adjacency matrix, creating explicit semantic shortcuts that enhance GNN-based reasoning. Our experimental results on the widely-used WebQSP and CWQ datasets confirm the effectiveness of our approach, showing that it not only achieves a superior trade-off between subgraph size and answer coverage but also consistently outperforms state-of-the-art baselines, particularly on complex multi-hop questions, while producing explicit and interpretable reasoning chains that directly address the ‘black-box’ nature of many neural KBQA systems.

CRedit authorship contribution statement

Xinglin Zhang: Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation; **Zhenyun Deng**: Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation; **Wen-tao Deng**: Writing – review & editing; **Lin Liu**: Writing – review & editing; **Jiuyong Li**: Writing – review & editing; **Hong-Yu Zhang**: Writing – review & editing; **Xiaoxia Li**: Writing – review & editing; **Debo Cheng**: Writing – review & editing; **Zaiwen Feng**: Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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